

Challenge archive: SekaiCTF 2023 - RandSubWare

deuterium — Event date: 2023-08-25

Challenge event began 2023-08-25; article publication date not recorded.

Canonical HTML: <https://deut-erium.github.io/challenges/sekaictf-2023/randsubware/>

Abstract

SekaiCTF 2023 - Cryptography.

By deut-erium.

Description excerpt

“RandSubWare” - standing for ‘Random Substitution Warefare Challenge’. This term encompasses the idea of breaking a substitution permutation network through the use of randomized and strategic attacks on the substituted boxes, engaging in a battle against the complexity and obscurity of the encryption.

Read this challenge as [plain text](#) or [JSON](#).

Browser practice

RandSubWare terminal

Practice only. Rewards are public dummy values, not event flags. Completion is saved separately on this device when storage is available. Nothing runs until you choose Start.

Version

Browser practice needs JavaScript and module Worker support.

Program output

Input lines

Enter sends; Shift+Enter adds a line. Input is not echoed in program output. Stop interrupts computation; Reset clears the session without starting another.

Browser limits and console API

Limits use UTF-8 bytes: 2 MiB per input line, 4 MiB and 4096 lines per batch (including separators), 8 MiB output per operation, and a 1 MiB visible transcript. Earlier output is replaced with a truncation notice. Setup progress is separate, capped at 1024 characters and four updates per second. Original challenge deadlines still apply after setup; there is no extra setup deadline.

The console API is `window.challengePractice: start(), send(stringOrLines), stop(), and reset()`. Select the version above before starting. Start and Send resolve at the next input prompt or program exit with `{output, state, solved}`; output contains only that operation's program text. Concurrent operations are rejected. Stop, Reset, deadlines and Worker failures reject in-flight operations; terminal errors carry partial output in `error.result`.

Strings accept LF or CRLF; a final newline ends the last line without adding another blank line. An empty string sends one blank line. Arrays contain literal lines without LF. Unused lines are discarded if the program exits. Reset keeps saved completion. Public dummy rewards do not unlock the original answer checker.

JavaScript is disabled. This terminal cannot start; its controls remain disabled.

Files

These files describe the original interactive service. The browser practice above is local; no TCP endpoint or retired remote service is connected.

- [chall.py](#)

The original remote service is retired. Browser practice runs locally with the public dummy reward practice—local dummy reward"; it does not award an event solve.

Download

```
mkdir -p sekaictf-2023-ransubware
cd sekaictf-2023-ransubware
curl --fail --location --output challenge.txt \
  "https://deut-erium.github.io/challenges/sekaictf-2023/ransubware/challenge.txt"
curl --fail --location --output challenge.json \
  "https://deut-erium.github.io/challenges/sekaictf-2023/ransubware/challenge.json"
curl --fail --location --output chall.py \
  "https://deut-erium.github.io/assets/challenges/sekaictf-2023-ransubware/chall.py"
```

Sources and solutions (spoilers)

- [Full archive](#) (may contain flags).
- [Solution](#)
- [Original description or source](#)
- [Event credits](#)
- [CC BY-NC-SA 4.0 \(description\)](#); [AGPL-3.0-or-later \(source\)](#).
- [Local AGPL-3.0 license](#)
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[All challenges](#) · [CTF tutorials](#)

| Interactive controls are omitted from this PDF. See the canonical HTML version.